

Motivation

Physical laws are often derived from first principles. When active terms are unknown, **data-driven equation discovery** offers a complementary route: infer a sparse governing equation directly from observations.

Research question and benchmark

Can sparse regression recover a Fokker-Planck equation directly from density data, including noisy trajectories?

Our benchmark is the Ornstein-Uhlenbeck process and its conservative density equation:

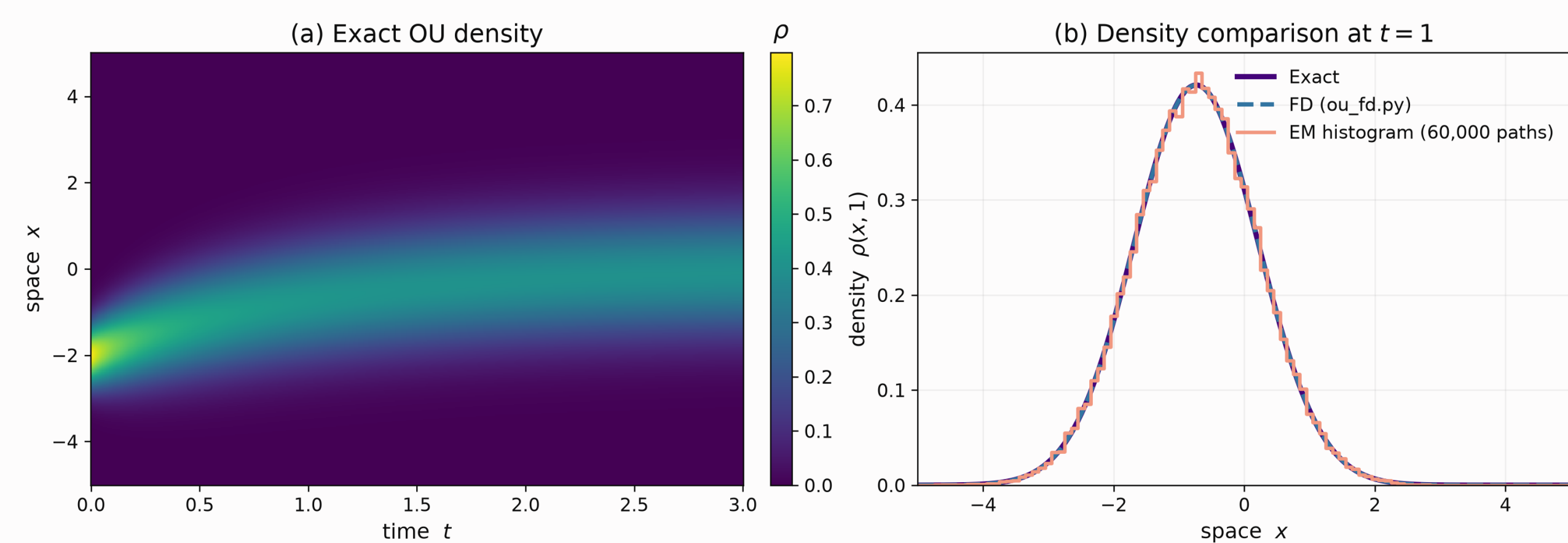
$$dX_t = -\kappa X_t dt + \sqrt{2D} dW_t, \quad \rho_t = \kappa \partial_x(x\rho) + D\rho_{xx}.$$

With $\kappa = D = 1$, $(x, t) \in [-5, 5] \times [0, 3]$ and $X_0 \sim \mathcal{N}(-2, 0.5^2)$, the true support is $\{\partial_x(x\rho), \rho_{xx}\}$ with coefficients $(1, 1)$.

Because both the dynamics and density are known, departures from $(1, 1)$ isolate errors due to data generation, numerical differentiation, library design or sparse regression.

Density data used

Aspect	Exact	Finite difference	Euler-Maruyama
Construction	analytic OU density	numerical FPE solution	60,000 paths + histogram
Main uncertainty	none (reference)	space-time discretisation	time stepping, sampling and binning



The three sources agree visually, but small bin-to-bin fluctuations in the EM histogram are strongly magnified by pointwise numerical derivatives.

Sparse identification of the PDE

Ordinary SINDy solves a sparse regression problem,

$$\mathbf{b} \approx \Theta \xi, \quad \mathbf{b} = \text{vec}(\rho_t),$$

using sequentially thresholded least squares (STLSQ). Each column of Θ represents a candidate PDE term; a representative strong-form library is

$$\rho, \rho_x, \partial_x(x\rho), \partial_x(x^2\rho), \rho_{xx}, \rho_{xxx}, \dots$$

Noise: if $\hat{\rho} = \rho + \varepsilon$, finite differences amplify noise approximately as ε/h for first derivatives and ε/h^2 for second derivatives.

Identifiability: larger libraries are not automatically better. The identity $\partial_x(x\rho) = \rho + x\rho_x$ creates dependent columns and non-unique coefficients. Prefer a compact conservative library; otherwise compare algebraically equivalent PDEs.

Weak SINDy: differentiate the test function, not the data

On each randomly sampled space-time subdomain Ω_k , integrate against a smooth compactly supported test function ϕ_k and transfer derivatives by parts:

$$\int_{\Omega_k} \rho_t \phi_k = - \int_{\Omega_k} \rho (\phi_k)_t, \quad \int_{\Omega_k} q_x \phi_k = - \int_{\Omega_k} q (\phi_k)_x,$$

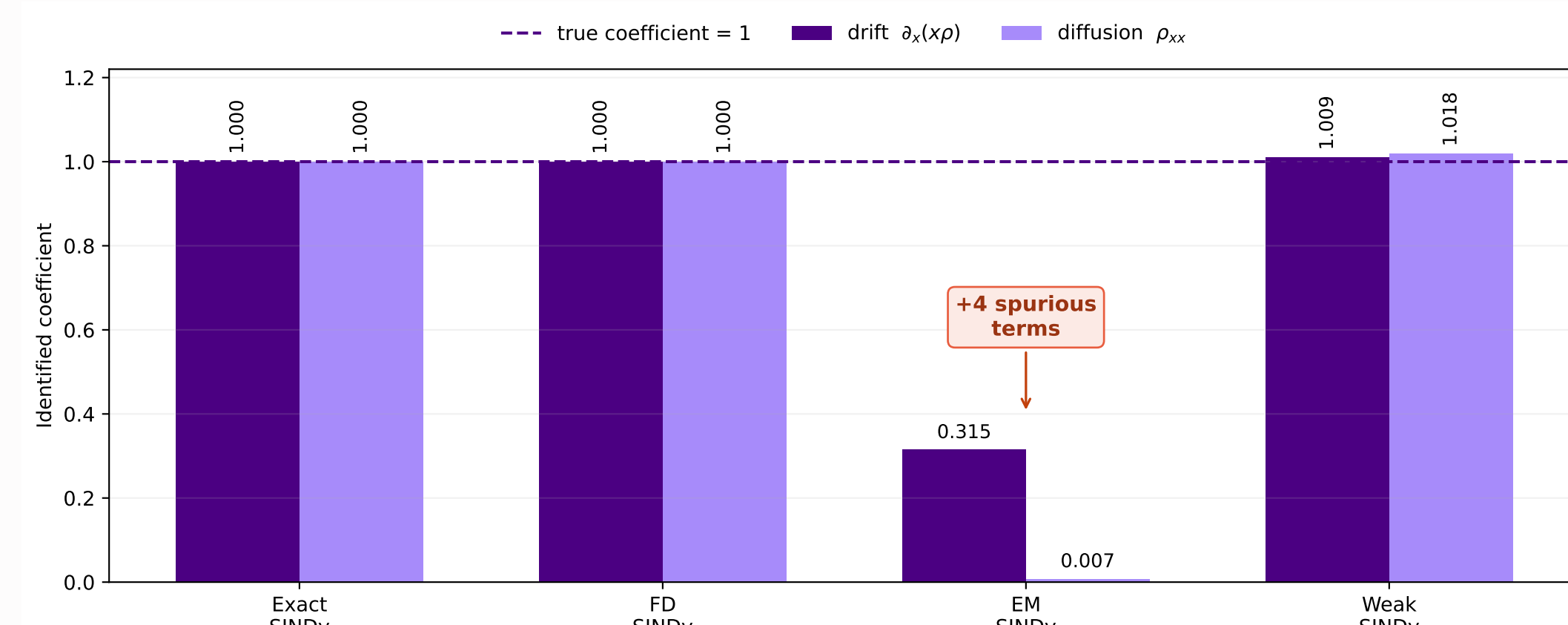
$$\int_{\Omega_k} \rho_{xx} \phi_k = \int_{\Omega_k} \rho (\phi_k)_{xx}.$$

Boundary terms vanish by construction. The observed density appears only inside local weighted integrals, so noise is averaged rather than differentiated.

Library 1	five selected columns: $\partial_x(x\rho)$, $\partial_x(x^2\rho)$, $\partial_x(x\rho^2)$, $\partial_x(x^2\rho^2)$, ρ_{xx}
Library 2	all 15 columns: five fields with derivative orders 0, 1, 2

Each subdomain supplies one weak equation: K sets the number of equations, while width sets the local averaging scale.

Weak integration restores coefficient recovery



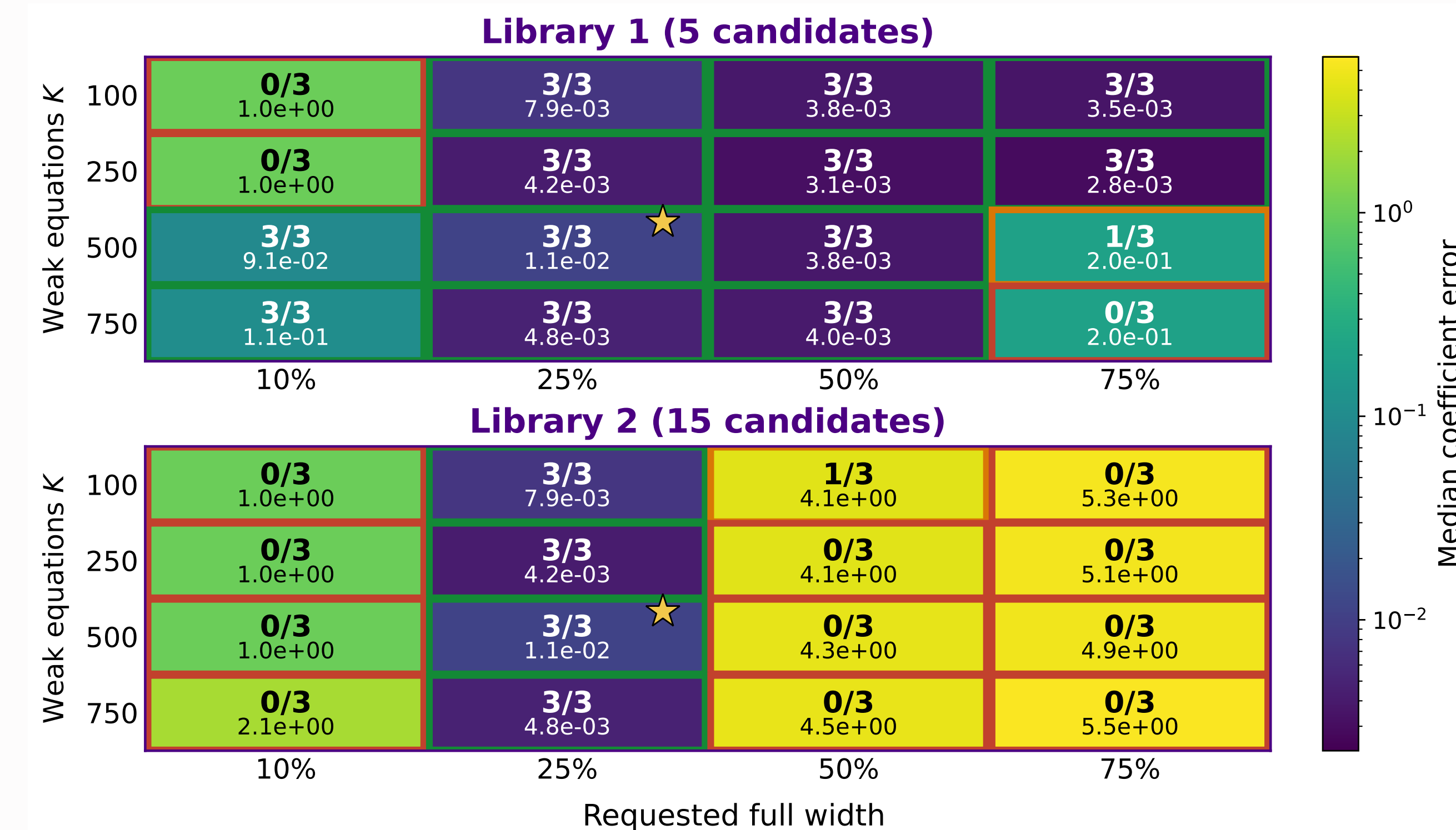
SINDy + Exact/FD	SINDy + EM	Weak SINDy + EM
correct support	diffusion nearly lost	correct support
coefficients close to 1	four spurious terms	coefficients close to 1

At $K = 500$ and 25% requested full width, both weak libraries recover the correct support with $\hat{\kappa} = 1.009$ and $\hat{D} = 1.018$.

K -width sensitivity across weak-subdomain seeds

On the fixed EM density, we sweep $K = 100, 250, 500, 750$ and requested full widths 10%, 25%, 50%, 75% using seeds 319, 802 and 2002. Cells report correct-support count out of three and median coefficient error; stars mark the camera-error setting.

25% is the safest common choice: both libraries recover the true support for every K and seed. At 75%, Library 2 fails in all three seeds despite a median residual of only 5.3×10^{-4} , so a small residual does not certify the correct PDE.

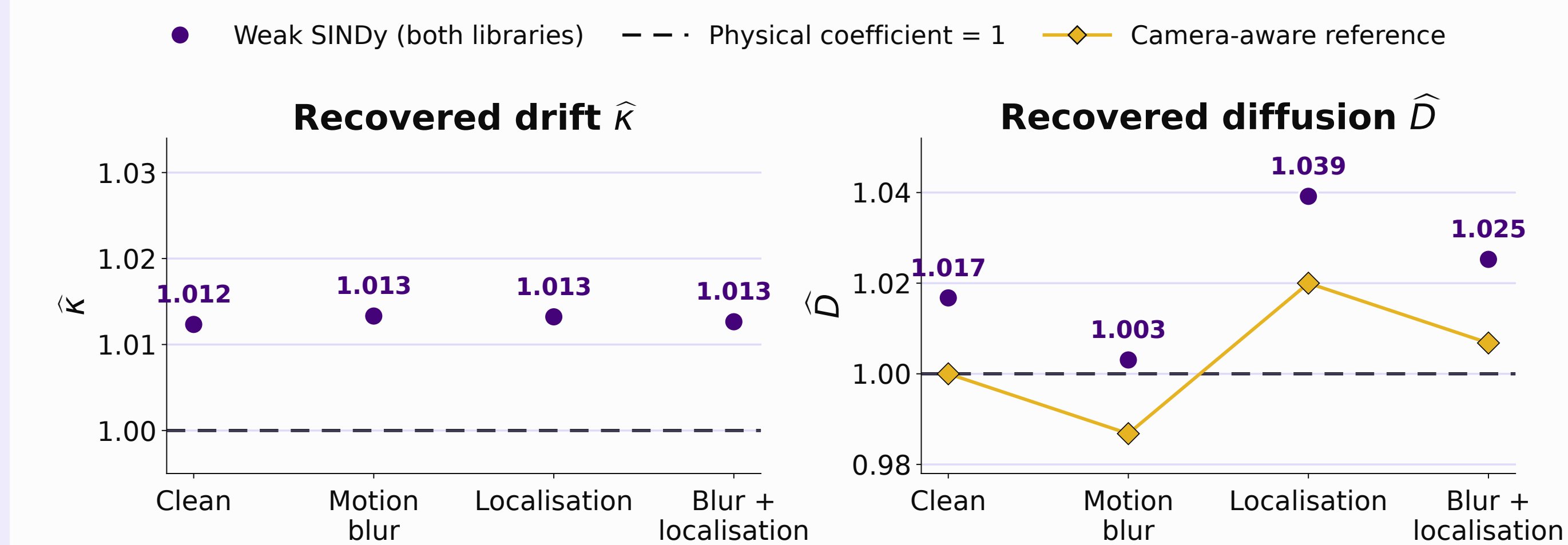


Robustness to camera errors

The same 60,000 OU paths are re-observed under four camera conditions:

Clean	midpoint positions
Motion blur	exposure average $X_{m,\alpha}^{\text{blur}} = \frac{1}{\Delta} \int_{t_\alpha}^{t_\alpha+\Delta} X_m(s) ds$, $\Delta = 0.04$
Localisation	additive Gaussian noise $X^{\text{loc}} = X^{\text{clean}} + \epsilon$ with $\epsilon \sim N(0, \sigma_{\text{loc}}^2)$
Both	motion blur + localisation

Fit: 60,000 paths; $K = 500$; 25% width; both libraries.



Across all conditions, both libraries recover exactly $\partial_x(x\rho)$ and ρ_{xx} ; camera-aware coefficient error is 1.47%–1.63%.

Camera bias remains: camera errors leave the selected PDE terms unchanged but shift the effective diffusion coefficient. Weak SINDy reduces derivative-amplified noise; it does not correct the camera observation model.

References

- [1] S. L. Brunton, J. L. Proctor and J. N. Kutz. Sparse identification of nonlinear dynamics. *PNAS*, 113:3932–3937, 2016.
- [2] S. H. Rudy, S. L. Brunton, J. L. Proctor and J. N. Kutz. Data-driven discovery of PDEs. *Science Advances*, 3:e1602614, 2017.
- [3] D. A. Messenger and D. M. Bortz. Weak SINDy for PDEs. *J. Computational Physics*, 443:110525, 2021.
- [4] A. J. Berglund. Statistics of camera-based single-particle tracking. *Physical Review E*, 82:011917, 2010.